

Amro Hybrid Constructive Generation System – Prime Constructor Enumeration

Definition:

A hybrid engineering system for generating ultra-fast prime candidate filters for numbers with millions of digits.

It relies on behavioral digit-construction instead of slow operations such as division or square-root extraction.

Important note: any number rejected by any constructor does not continue to the next stage and is considered composite immediately.

1) Stage One: Initial Formation – Amro Generator

Base set: All even numbers: 4, 6, 8, 10, 12, ...

Formation process: Add constant value 3 to every base number

Output: Always an odd prime-candidate: $4 \rightarrow 7$, $6 \rightarrow 9$, $8 \rightarrow 11$

2) Stage Two: Fast Behavioral Construction ($O(k)$)

Purpose: eliminate most composites extremely quickly before any heavy operations.

Constructors are applied sequentially.

If one constructor rejects, the number stops and is composite immediately.

A) Constructor Coefficient Law (k-Law):

Find smallest k such that:

- $10k - 1$ divisible by $P \rightarrow$ ADDITION rule ($a + kb$)
- $10k + 1$ divisible by $P \rightarrow$ SUBTRACTION rule ($a - kb$)

Example for $P = 59$:

Testing subtraction:

$k=1 \rightarrow 11$ ~~X~~

$k=6 \rightarrow 61$ ~~X~~

$k=12 \rightarrow 121$ ~~X~~

$k=18 \rightarrow 181$ ~~X~~

$k=35 \rightarrow 351X$

Testing addition:

$k=1 \rightarrow 9X$

$k=6 \rightarrow 59 \checkmark$ divisible

Final:

Constructor = 59

$k = 6$

Rule = ADDITION $\rightarrow R$ reduces as $a + 6b$

Behavioral Constructor Table ($3 \rightarrow 43$):

$3 \rightarrow$ digit sum rule

$5 \rightarrow$ unit digit rule

$11 \rightarrow$ alternating subtraction/addition of grouped digits

$7 \rightarrow a - 2b$

$13 \rightarrow a + 4b$

$17 \rightarrow a - 5b$

$19 \rightarrow a + 2b$

$23 \rightarrow a + 7b$

$29 \rightarrow a + 3b$

$31 \rightarrow a - 3b$

$37 \rightarrow a - 11b$

$41 \rightarrow a - 4b$

$43 \rightarrow a + 13b$

Rejection at any constructor = composite immediately.

3) Stage Three — Confirmatory Deep Check

Used only for numbers surviving all fast constructors.

- Sub-segment sequential check
- Miller–Rabin probabilistic test
- Large-prime family divisibility check (47, 53, ...)

Mathematical Proof of k-Law

Let $R = 10a + b$

Addition rule proof:

$10k \equiv 1 \pmod{P} \rightarrow R \text{ divisible by } P \Leftrightarrow a + kb \text{ divisible by } P$

Subtraction rule proof:

$10k \equiv -1 \pmod{P} \rightarrow R \text{ divisible by } P \Leftrightarrow a - kb \text{ divisible by } P$

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